

CHAPTER 9 : DISASTER MANAGEMENT

Topic:-

Make a collection of news, photos and cuttings about landslides and rift collapse.

Introduction:-

Nature is magnificent, but it can also unleash destructive natural disasters. A landslide is the rapid downward movement of rock masses, soil, and debris sliding down a steep hillside or mountain slope under the force of gravity. A rift collapse (or ground subsidence) occurs when underground support collapses, causing cracks and sudden sinkholes to open on roads and farmland.

In Maharashtra, the Western Ghats (Sahyadri mountain range) experience heavy monsoon rainfall every year. Hill stations and mountain villages in Raigad, Pune, Satara, and Ratnagiri districts are highly prone to devastating landslides. Famous historic tragedies like the Malin village disaster (Pune, 2014) and the Irshalwadi disaster (Raigad, 2023) buried entire villages in mud while people were sleeping. In this disaster management project, we created a scrapbook collection of newspaper reports, photos, and clippings to study landslides.

❖ Main Causes of Landslides:-

- **Natural Factors:** Torrential monsoon rainfall, seismic earthquakes, and steep natural slopes.
- **Man-Made Factors:** Deforestation, illegal tree cutting on hillsides, and reckless blasting for highway building.

Aim & Scrapbook Collection Plan:-

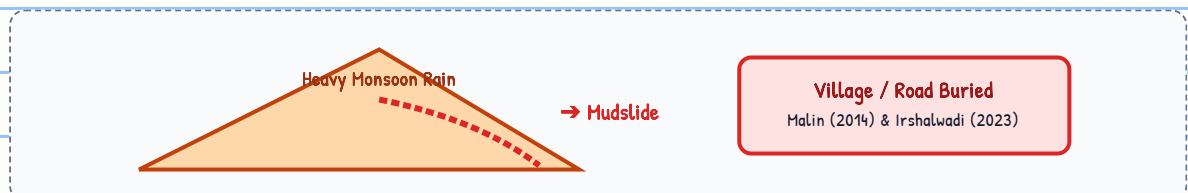
Aim: To collect newspaper articles, photographs, and magazine clippings of landslides and rift collapses in Maharashtra and India, and analyze their causes, destruction, and disaster mitigation measures.

❖ Objectives of the Project:-

- To document major landslide disasters reported in newspapers over recent years.
- To identify early warning signals of hillside movement (cracks on walls and trees leaning).
- To study rescue operations led by NDRF, SDRF, and local youth volunteers.

❖ Materials Required:-

- Old newspapers (Maharashtra Times, Lokmat, Times of India, Indian Express).
- Scrapbook album, scissors, glue stick, sketch pens, and highlighter markers.



Procedure & Newspaper Scrapbook Compilation:-

We collected and cataloged disaster clippings following this organized process:

- **Step 1: Gathering Old Papers:** Collected old Marathi and English daily newspapers published during July and August monsoon months from libraries and neighbors.
- **Step 2: Cutting Articles & Photos:** Cut out headline reports, photos of damaged roads, collapsed mountain slopes, and aerial photos of rescue teams.
- **Step 3: Chronological Sorting:** Organized articles by disaster events, dating from the historic 2014 Malin landslide to the recent 2023 Irshalwadi disaster and Mumbai-Goa highway rift collapses.
- **Step 4: Writing Explanatory Captions:** Glued each clipping on scrapbook pages and wrote brief notes detailing: Date, Location, Casualties, Probable Trigger, and Government Action.

❖ Early Warning Signs of an Impending Landslide:-

- New vertical cracks appearing on hill slopes, house floors, and concrete foundations.
- Telephone poles, electric posts, and tall trees leaning downhill.
- Sudden muddy brown water gushing out of hillside springs that normally run clear.



Landslide Case Studies Table:-

The table below documents the major landslide events recorded in our scrapbook:

<i>Disaster Event</i>	<i>Location & Date</i>	<i>Triggering Factor</i>	<i>Destruction Caused</i>	<i>Rescue Force Deployed</i>
Malin Landslide	Ambegaon, Pune (30 July 2014)	108 mm heavy rain in 24 hrs + hill flattening	Entire village buried; 151 precious lives lost	NDRF battalions, heavy excavators, sniffer dogs
Taliye Village Slide	Mahad, Raigad (22 July 2021)	Extreme torrential flood cloudburst	35 houses buried under 15 feet of red mud	Indian Army, Navy helicopters, NDRF
Irshalwadi Hill Collapse	Khalapur, Raigad (19 July 2023)	Steep tribal hill slope saturation	Remote settlement wiped out; 84 casualties	NDRF jawans carrying stretchers on foot
Kamshet Tunnel Rift	Mumbai-Pune Expressway (Monsoon 2023)	Rock weathering & heavy vibrations	Highway blocked by giant boulders for 2 days	Highway Police, mountain boulder netting teams

Scientific Analysis & Discussion:-

Examining the newspaper reports revealed why mountain slopes collapse during rains:

- **How Water Saturates Soil:** Dry mountain soil is held together by tree roots and friction. When 100+ mm of rain falls in a few hours, water soaks into porous soil layers. Heavy waterlogged soil becomes liquid-like mud, destroying internal friction and causing the entire hillside to slide down like a river of concrete.
- **The Deadly Role of Deforestation:** Tree roots act like natural biological steel reinforcement nets that bind hillside topsoil. When trees are cut down for tourist resorts or road widening, the protective anchor is lost, making slopes 10 times more vulnerable!
- **Unplanned Hill Terrace Digging:** In Malin, JCB machines had flattened hill slopes for commercial potato farming, cutting into the natural toe support of the hill.

❖ Why Nighttime Landslides Are Deadlier:-

Most major slides occurred between 3:00 AM and 5:00 AM while villagers were sound asleep, giving them zero time to escape.

Prevention & Disaster Mitigation Measures:-

Disaster management experts recommend proactive engineering and ecological solutions:

- **1. Afforestation of Vulnerable Slopes:** Planting deep-rooting native trees (Bamboo, Vetiver grass, Banyan) along steep hill slopes binds loose soil naturally.
- **2. Steel Geo-Netting along Highways:** Fixing heavy high-tensile steel wire mesh blankets over rock faces along mountain highways catches falling boulders safely.
- **3. Proper Hill Drainage Channels:** Digging concrete catch-water drains along hill contours diverts rushing rainwater away from populated settlements below.
- **4. Early Evacuation Protocols:** When weather agencies predict red alert rain exceeding 200 mm, hill tribal families should be immediately moved to temporary shelter camps.

❖ Do's and Don'ts During Landslide Warnings:-

- **DO:** Move quickly away from the path of mudflows to higher ground immediately.
- **DON'T:** Never seek shelter under highway bridges or near steep riverbanks during torrential rain.

What We Learned & Conclusion:-

- Landslides are rapid downward soil collapses triggered by torrential rain, steep terrain, and deforestation.
- Man-made disturbances (hill cutting, tree felling, road blasting) drastically increase landslide risks.
- Early warning signs—cracks in walls, tilted trees, muddy springs—must never be ignored.
- Steel wire netting, reforestation, and timely monsoon evacuation save hundreds of human lives.

Conclusion:-

Creating this newspaper scrapbook on landslides and rift collapses was a moving educational experience. It taught us that natural disasters are closely linked to how humans treat mountain ecosystems. By respecting nature, preserving hill forests, and implementing smart geotechnical disaster planning, we can protect vulnerable hill communities from catastrophic mudslides.

References:-

- Maharashtra State Board Class 8 General Science Textbook (Chapter 9: Disaster Management).
- Geological Survey of India (GSI) — Landslide Hazard Zonation Atlas of Maharashtra.
- Newspaper archives: The Times of India, Lokmat, and Maharashtra Times (2014–2023).